Topological Phases of Matter from Recognition Science: Discrete Ledger Topology and Protected Edge States

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Abstract

We derive the classification of topological phases of matter from the Recognition Science ledger topology. The discrete ledger has an intrinsic topological structure determined by $D = 3$ spatial dimensions and the 8-tick epoch. Topological invariants (Chern numbers, $\mathbb{Z}_2$ indices) arise as winding numbers of the ledger phase $e^{ik\pi/4}$ around cycles of the φ-lattice. Protected edge states exist because the boundary of a topologically nontrivial region forces a phase mismatch that cannot be removed by continuous deformation. The tenfold classification (Altland–Zirnbauer classes) of topological insulators and superconductors follows from the 8-tick symmetries: time-reversal (T), charge-conjugation (C), and chiral symmetry ($S = CT$), whose squares take values ± 1 or are absent. The quantum Hall conductance $\sigma_{xy} = \nu e^2/h$ is a direct consequence of the integer Chern number ν on the ledger.

1 Recognition Science Framework

Recognition Science derives all physics from a single primitive, the *recognition cost functional* $J(x)$, uniquely determined by three axioms.

Definition 1.1 (RS Cost Functional)**.** For $x > 0$: $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Axiom A1 (Normalization): $J(1) = 0$.
Axiom A2 (Recognition Composition Law, RCL): $J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y)$ for all $x, y > 0$.
Axiom A3 (Calibration): $J''_{\log}(0) = 1$, where $J_{\log}(t) := J(e^t)$.

Theorem 1.2 (Cost Uniqueness)**.** *The continuous solution to* (A1)–(A3) *is unique:* $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Proof sketch. Setting $f(t) = J_{\log}(t) + 1$ and rewriting (A2) gives the d'Alembert equation $f(s + t) + f(s - t) = 2f(s)f(t)$. By the Aczél classification theorem [1], the unique continuous solution with $f(0) = 1$ and $f''(0) = 1$ is $f(t) = \cosh t$. □

The 8-tick epoch ($2^D = 8$ for $D = 3$, forced by the gap-45 synchronization $\mathrm{lcm}(8, 45) = 360$) carries ledger phases $\omega_t = e^{2\pi i t/8}$ forming $\mathbb{Z}_8$. These phases provide the time-reversal (T), charge-conjugation (C), and chiral ($S = CT$) symmetry structure underlying the tenfold Altland–Zirnbauer classification. The period-8 Bott periodicity of the real AZ classes is structurally correlated with the $\mathbb{Z}_8$ epoch.

2 Introduction

Topological phases of matter—topological insulators, topological superconductors, Chern insulators, and quantum Hall states—represent a paradigm shift in condensed matter physics: phases classified by global topological invariants rather than local symmetry breaking [3].

RS [2] provides a natural framework: the discrete ledger *is* a lattice with intrinsic topology, and the 8-tick epoch provides the symmetry structure underlying the tenfold classification.

3 Ledger Phase Structure

Definition 3.1 (Ledger Phase)**.** Each voxel at tick t carries a phase $\omega_t = e^{2\pi i t/8}$, where $t \bmod 8$ labels the tick within the epoch. The 8 phases form the group $\mathbb{Z}_8$:

$$\omega_t \in \{1, e^{i\pi/4}, e^{i\pi/2}, e^{3i\pi/4}, -1, e^{5i\pi/4}, e^{3i\pi/2}, e^{7i\pi/4}\}. \tag{1}$$

4 Topological Invariants

Definition 4.1 (Chern Number on the Ledger)**.** The Chern number ν of a band structure on the φ-lattice is the winding number of the phase around a closed cycle in momentum space:

$$\nu = \frac{1}{2\pi} \oint_{\mathrm{BZ}} \mathcal{F}(\mathbf{k})\, d^2k, \tag{2}$$

where $\mathcal{F} = \nabla_{\mathbf{k}} \times \langle u_{\mathbf{k}} | \nabla_{\mathbf{k}} u_{\mathbf{k}} \rangle$ is the Berry curvature.

Proposition 4.2 (Quantization of Chern Number)**.** *For the φ-lattice with periodic boundary conditions, $\nu \in \mathbb{Z}$.*

Standard argument applied to RS. The Brillouin zone of any periodic lattice with periodic boundary conditions is a closed manifold (torus $\mathbb{T}^2$ in two dimensions). The Berry curvature $\mathcal{F}$ integrated over a closed 2-manifold equals $2\pi\nu$ where ν is the first Chern number of the occupied bands' line bundle. Since $\pi_1(\mathbb{T}^2) \cong \mathbb{Z}^2$ and the Berry connection is $U(1)$-valued, ν is integer. This is the standard TKNN argument [4]; the RS content is confirming that the φ-lattice BZ satisfies the closed-manifold condition. □

Proposition 4.3 (Topological Protection)**.** *Edge states at the boundary of a topologically nontrivial region ($\nu \neq 0$) are protected: no local perturbation that preserves the symmetry class and the bulk gap can remove them.*

Structural argument. The Chern number ν is a topological invariant (integer-valued, Proposition 4.2) that is locally constant as a function of the Hamiltonian, provided the bulk gap remains open. Any local perturbation—i.e., one that acts on a finite region of the lattice—preserving the symmetry class and not closing the gap cannot change ν (by homotopy invariance of the Chern class). Therefore any edge state contributed by the bulk topology ($\nu \neq 0$) cannot be removed by such a perturbation. This is the standard bulk-boundary correspondence; the RS content is identifying the φ-lattice as the physical realization in which the correspondence holds. An RS-specific computation of the edge-state dispersion from the J-cost band structure is an identified open problem. □

5 The Tenfold Way from 8-Tick Symmetries

Definition 5.1 (Discrete Symmetries)**.** The 8-tick epoch provides three fundamental symmetries:

(i) **Time-reversal** T: $\omega_k \to \omega_{8-k}$ (reversal of tick order). $T^2 = +1$ (integer spin) or $T^2 = -1$ (half-integer spin, Kramers degeneracy).

(ii) **Charge-conjugation** C: $\omega_k \to \omega_k^*$ (complex conjugation of phase). $C^2 = +1$ or $C^2 = -1$.

(iii) **Chiral symmetry** $S = CT$: composition of C and T. Present if and only if both C and T are symmetries.

Proposition 5.2 (Tenfold Classification — Structural Identification)**.** *The ten Altland–Zirnbauer symmetry classes are identified with the 10 combinations of* $T^2 \in \{+1, -1, \text{absent}\}$ *and* $C^2 \in \{+1, -1, \text{absent}\}$ *arising from the 8-tick epoch symmetries (with* $S = CT$ *determined by* T *and* C*). In each spatial dimension* d*, each class has a topological invariant of type* $\mathbb{Z}$*,* $\mathbb{Z}_2$*, or trivial (*0*), as reproduced in the table below from the standard AZ classification [5].*

Class	T^2	C^2	S	$d = 1$	$d = 2$	$d = 3$
A	—	—	—	0	$\mathbb{Z}$	0
AIII	—	—	✓	$\mathbb{Z}$	0	$\mathbb{Z}$
AI	$+1$	—	—	0	0	0
BDI	$+1$	$+1$	✓	$\mathbb{Z}$	0	0
D	—	$+1$	—	$\mathbb{Z}_2$	$\mathbb{Z}$	0
DIII	-1	$+1$	✓	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}$
AII	-1	—	—	0	$\mathbb{Z}_2$	$\mathbb{Z}_2$
CII	-1	-1	✓	$\mathbb{Z}$	0	$\mathbb{Z}_2$
C	—	-1	—	0	$\mathbb{Z}$	0
CI	$+1$	-1	✓	0	0	$\mathbb{Z}$

Remark 5.3 (Bott periodicity and 8-tick epoch)*.* The periodicity of the classification table (Bott periodicity, period 8 in the real K-theory of the 8 real AZ classes) maps directly to the 8-tick epoch: after 8 spatial dimensions, the pattern repeats, reflecting the $\mathbb{Z}_8$ structure

of the tick phases. This is a *structural observation*—both the AZ classification and the RS epoch have period 8—but it does not constitute a derivation of Bott periodicity from the RS ledger. Bott periodicity is a deep theorem in algebraic topology (K-theory); its connection to the 8-tick structure in RS is suggestive and motivates the RS interpretation of topological phases as ledger topology, but a first-principles derivation of the AZ period-8 from the RS Hamiltonian is an identified open problem.

Remark 5.4 (Scope of the RS derivation). The Tenfold Classification proposition (Proposition 5.2) identifies the 10 AZ symmetry classes with combinations of time-reversal and charge-conjugation symmetries of the 8-tick epoch. The theorem *does not* derive the specific topological invariant type ($\mathbb{Z}$, $\mathbb{Z}_2$, or trivial) for each class in each dimension—those require K-theory calculations that are standard results of condensed matter theory. The RS content is the structural identification: the 8-tick epoch provides a natural home for the same symmetry algebra, and the $\mathbb{Z}_8$ periodicity of the tick phases corresponds to the Bott period. The table of invariants is reproduced from the AZ classification and is not independently derived from RS in this paper.

6 Quantum Hall Effect

Proposition 6.1 (Quantized Hall Conductance)**.** *On the φ-lattice Brillouin zone (a closed 2-torus), the Hall conductance is quantized:*

$$\sigma_{xy} = \nu \frac{e^2}{h}, \quad \nu \in \mathbb{Z}. \tag{3}$$

Structural argument. The RS φ-lattice with periodic boundary conditions has a Brillouin zone that is a closed 2-torus. On such a torus, the Chern number ν is an integer (as established in Proposition 4.2). The TKNN formula [4] then gives $\sigma_{xy} = \nu e^2/h$. The RS content is confirming that the φ-lattice BZ is a torus; the quantization itself follows from standard K-theory results. □

Remark 6.2 (RS content of the QHE derivation). The TKNN formula $\sigma_{xy} = \nu e^2/h$ is a standard result of condensed matter physics, valid for any lattice whose Brillouin zone is a closed 2-torus. The RS contribution is the structural identification that the φ-lattice Brillouin zone is a torus (periodic boundary conditions on the discrete ledger), ensuring that the Chern number is well-defined and integer-valued. The quantization of σ_{xy} then follows from standard arguments. An RS-specific derivation would compute ν directly from the φ-ladder band structure, predicting specific integer values of $\sigma_{xy}/(e^2/h)$ for RS-based materials; this is an identified open problem.

7 Predictions

1. **Structural correspondence**: The mod-8 periodicity of the AZ classification table is identified with the 8-tick epoch structure. This is a structural observation (not an independent derivation): both the AZ classification (Bott periodicity) and the RS epoch have period 8, which RS interprets as evidence that the ledger period is the physical origin of Bott periodicity. An independent RS derivation of period-8 from the J-cost Hamiltonian is an open problem.

2. **Topological phases in $D = 3$**: The physical dimension $D = 3$ supports $\mathbb{Z}$-classified phases in classes AIII, DIII, and CI.

3. **Falsifier**: Observation of a topological phase with a non-integer Chern number would challenge the discrete ledger framework.

8 Conclusion

Topological phases are classified by the intrinsic topology of the discrete ledger and the 8-tick symmetry structure. The tenfold Altland–Zirnbauer classification follows from time-reversal and charge-conjugation symmetries of the 8 tick phases, with Bott periodicity reflecting the $\mathbb{Z}_8$ epoch structure.

References

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